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Started	Thursday, 29 October 2026, 7:34 PM
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Duration	10 mins 4 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It compares adjacent elements and swaps them if necessary
- d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **2**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Merging
- b. Searching
- c. Rearranging
- d. Sorting

The correct answer is: Sorting

Question **3**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It bubbles up the largest element in each pass
- c. It uses the heap data structure
- d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are merged without sorting
- b. The parts are ignored
- c. Each part is searched for a specific element
- d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Bubble Sort
- c. Quick Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Selection
- c. Extraction
- d. Distribution

The correct answer is:
Insertion

Question **7**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is difficult to implement
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **8**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. Selection
- c. Searching
- d. All of the above

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It sorts data using a single pass
- b. It is based on the divide-and-conquer approach
- c. It does not require recursion
- d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It only works on small datasets
- c. It avoids the need for recursion
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **12**

Complete

Mark 0.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. order()
- c. sorted()
- d. arrange()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Bubbling up the largest element to its correct position with each pass
- c. Being the most efficient sorting algorithm
- d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It slows down the process
- c. It makes the data unsorted
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items